Text to Structured Data using Entity Relationship Triples and Knowledge Graph

Ananya Kasavajhala (2023113025)
Jayanth Raju (2023115016)
Tulasi Rushwik (2023111011)
Viswas Reddy (2023101008)

Introduction to NLP — Prof. Manish Shrivastava

Problem Statement

Task: Given a biography paragraph, generate a structured table (Wikipedia infobox)

> *"Leonard Shenoff Randle (born February 12, 1949) is a former Major League Baseball player. He was the first-round pick of the Washington Senators..."*

Attribute	Value
name	Leonard Shenoff Randle
birth_date	February 12, 1949
occupation	baseball player
team	Washington Senators

We construct an intermediate **knowledge graph** and then flatten it into a table.

Motivation

- Wikipedia has **millions of articles** but many lack structured infoboxes
- Structured data is critical for search engines, QA systems, and knowledge bases
- Manually building infoboxes does not scale to millions of entries
- Existing relation extraction methods often need a **predefined schema**
  - We wanted to compare generative (seq2seq) vs. discriminative (span + classify) approaches
- An intermediate knowledge graph gives us a **checkpoint to inspect and fix errors** before the final table
  - The pipeline is modular — each stage can be improved independently

Related Work

Work	Contribution
REBEL (Huguet Cabot & Navigli, 2021)	End-to-end relation extraction via seq2seq; 200+ Wikidata relations
GLiNER (Zaratiana et al., 2023)	Generalist NER with flexible entity labels; no retraining needed
LUKE (Yamada et al., 2020)	Entity-aware transformer; learns entity embeddings alongside tokens
Sentence-BERT (Reimers & Gurevych, 2019)	Dense sentence embeddings; we use it for relation canonicalization
WikiBio (Lebret et al., 2016)	728K biography text-table pairs; standard benchmark
Text-to-Table (Wu et al., 2022)	Transformer-based text-to-table; similar evaluation methodology

Dataset: WikiBio

- **728,321** biographies from English Wikipedia (enwiki-20150901 dump)
- Pre-split: **~582K** train / **~72K** test / **~72K** valid

File	Contents
`.sent`	Biography paragraph (free text)
`.box`	Infobox attributes (tab-separated key-value)
`.title`	Wikipedia article title
`.nb`	Number of sentences per paragraph

- Test set used for evaluation: **36,415** unique records, **140,249** gold attribute-value triples
- Common attributes: `birth_date`, `name`, `nationality`, `occupation`, `birth_place`, `position`

Pipeline Overview

Input → Biography paragraph **Output** → Structured attribute-value table

Stage	Description
1. Preprocessing	Clean WikiBio tokens, sentence segmentation
2. Triple Extraction	Extract (subject, relation, object) triples
3. Knowledge Graph	Group sentence-level triples per person
4. KG → Table	Normalize relations, deduplicate, assemble table

Three approaches for **Stage 2** (triple extraction):

Approach	Method
A	REBEL-only (base + fine-tuned)
B	Hybrid: GLiNER entity grounding + REBEL relations
C	BERT span extraction + LUKE relation classification + SBERT canonicalization

Text Preprocessing

WikiBio has dataset-specific tokens that need cleanup:

WikiBio Token	Replacement
`-lrb-` / `-rrb-`	`(` / `)`
`-lsb-` / `-rsb-`	`[` / `]`
`-lcb-` / `-rcb-`	`{` / `}`

Steps applied:

1. Replace WikiBio bracket tokens
2. Unicode NFKC normalization
3. Remove citation markers — `[1]`, `[citation needed]`, etc.
4. Fix punctuation spacing, strip extra whitespace
5. Sentence segmentation via spaCy (`en_core_web_trf`)

Before: `randle -lrb- born february 12 , 1949 -rrb- is a former ...`
After: `Randle (born February 12, 1949) is a former ...`

Approach A: REBEL-Only

REBEL (`Babelscape/rebel-large`): Seq2seq model that generates triples directly

Input: biography sentence → **Output:** `<triplet> subject <subj> object <obj> relation`

Design choices:

- **Pronoun coreference** — `he` / `she` → replaced with the person's title
- **Concatenated relation splitting** — REBEL can output merged relations like `"date of birth place of birth"`; split via greedy left-to-right match against seed vocabulary
- **Buffered processing** — adjacent sentences processed together for cross-sentence context
- Also **fine-tuned** REBEL on WikiBio infobox data converted into `<triplet>` format

REBEL Fine-tuning Details

Base model: `Babelscape/rebel-large`

Training data: WikiBio infoboxes parsed into REBEL `<triplet>` labels

Parameter	Value
GPUs	4 (Distributed Data Parallel)
Effective batch size	64 (2 × 4 GPUs × 8 accumulation)
Learning rate	3e-5
Max epochs	10
Early stopping	patience 5 on eval loss
Source / Target max length	512 / 256 tokens
FP16	Yes

Custom preprocessing script parsed indexed attributes, column vectors, sub-records, and removed filler tokens from WikiBio TSV infoboxes.

TODO: Fill actual training metrics — epochs completed, final train/eval loss

Approach B: Hybrid — GLiNER + REBEL

Idea: Use GLiNER2 for entity grounding, REBEL for relation extraction

GLiNER2 (`fastino/gliner2-base-v1`) identifies entities with custom labels:
`person`, `occupation`, `education`, `birth_date`, `birth_place`, `nationality`, `organization`, `award`, `notable_work`

Entity resolution step (the key addition):

- Match REBEL's short object outputs against GLiNER's full entity spans
- Example: REBEL outputs `"baseball"`, GLiNER detected `"major league baseball"` → resolved to the longer form
- Extra occupation triples inferred from GLiNER entity labels

Scale on test set:

44,144 sentences → 20,744 unique records → **124,784** triples after entity resolution

Approach C: BERT + LUKE + Sentence-BERT

Discriminative pipeline — **three models**, each handling a different subtask:

Step	Model	Task	Key Details
1. Span Extraction	`bert-base-uncased`	BIO token classification (`O`, `B-ATTR`, `I-ATTR`)	3 epochs, lr 3e-5, batch 16
2. Relation Classification	`studio-ousia/luke-base`	Classify each span's relation type	4 epochs, lr 2e-5, batch 16
3. Canonicalization	`all-MiniLM-L6-v2`	Cluster & normalize relation labels	Agglom. clustering, cosine threshold 0.25

- **Step 1** identifies *where* attribute values appear in the sentence
- **Step 2** input: `title: {title} [SEP] span: {span} [SEP] sentence: {sent}`
- **Step 3** groups similar relation strings (e.g. `"born in"` & `"place of birth"`) → picks the most frequent as canonical

BERT + LUKE: Training Data Preparation

Converting WikiBio infoboxes into supervised training data:

Component	Approach
Value alignment	Match each infobox value to biography text (exact substring first, fuzzy token F1 ≥ 0.6 as fallback)
BIO tagging	`B-ATTR` for first matched token, `I-ATTR` for continuation; sub-word tokens use `word_ids()`, continuation sub-words get `-100`
Relation labels	Each aligned span paired with its infobox attribute name
Noise filtering	Excluded: `image`, `caption`, `signature`, `logo`, `website`, `url`
Sub-field merging	`birth_date_1`, `birth_date_2` → `birth_date`

BERT + LUKE: Inference Pipeline

At inference, the three models are chained sequentially:

sentence + title → BERT Span Model → *list of spans* → LUKE Relation Model (per span) → *relation label* → SBERT Canonical Map → **output triples**

- Sentences where BERT predicts only `O` tags → no triples emitted
- Each extracted span is independently classified by LUKE
- Canonical relation map is pre-computed once and applied as a lookup at inference
- Final output per sentence: list of `(title, span_text, relation)` triples

Knowledge Graph → Table Conversion

KG Construction: Group sentence-level triples by person title; deduplicate; store as `{title → {relation → [values]}}`

KG → Table pipeline:

Phase	What it does
Anchor Detection	Match subject to biography person via alias set
Relation Normalization	80+ mapping rules; disambiguate date vs. place via entity types
Deduplication	Case-insensitive + substring absorption
Filtering	Drop noise: `instance_of`, `subclass_of`, `part_of`, etc.
Assembly	Order: name → birth → death → personal → career → other

Example dedup: `"baseball"` absorbed by `"major league baseball"`

Evaluation Methodology

Metric	Description
chrF	Character n-gram F-score (n = 1–6, β = 2.0, recall-weighted)
Triple P / R / F1	Fuzzy matching: similarity = 0.35 × relation_sim + 0.65 × value_sim

Fuzzy triple matching details:

- Value similarity = max(SequenceMatcher ratio, token F1)
- A predicted triple counts as matched if similarity ≥ **0.78**
- Greedy one-to-one alignment between predicted and gold triples

Gold standard: Constructed using WikiBio test set — **36,415** records, **140,249** triples

Results: BERT + LUKE Pipeline

Metric	Raw Relations	Canonicalized
chrF	**62.25**	60.38
Precision	**0.755**	0.748
Recall	**0.604**	0.598
F1	**0.671**	0.664
Matched Triples	84,689	83,839
Predicted Triples	112,129	112,129
Gold Triples	140,249	140,249

- **Coverage:** 34,854 / 36,415 test records (**95.7%**)
- Raw outperforms canonical slightly — the Sentence-BERT clustering (threshold 0.25) over-merges some distinct relation types that happen to be semantically close

Results: Cross-Approach Comparison

TODO: Fill in REBEL-only and hybrid pipeline metrics once evaluation completes

Approach	chrF	Precision	Recall	F1
REBEL-only (base)	*[TODO]*	*[TODO]*	*[TODO]*	*[TODO]*
REBEL-only (finetuned)	*[TODO]*	*[TODO]*	*[TODO]*	*[TODO]*
GLiNER + REBEL (hybrid)	*[TODO]*	*[TODO]*	*[TODO]*	*[TODO]*
BERT + LUKE (raw)	**62.25**	**0.755**	**0.604**	**0.671**
BERT + LUKE (canonical)	60.38	0.748	0.598	0.664

TODO: Observations and analysis comparing the three approaches

Qualitative Examples — Good Results

Input: "*Leonard Shenoff Randle (born February 12, 1949) is a former Major League Baseball player...*"

Attribute	BERT+LUKE Predicted	Gold
birth_date	february 12, 1949	✓
debutteam	washington senators	✓

Input: "*Philippe Adnot (born 25 August 1945 in Rhèges) is a member of the Senate of France...*"

Attribute	BERT+LUKE Predicted	Gold
birth_date	25 august 1945	✓
constituency	aube	✓
occupation	farmer	✓

Failure Cases & Error Analysis

1. Long input degeneration (Finetuned REBEL — Jack Reynolds):

> Output: `...1919 1919 1919 1919 1919 1919...` — repeating year tokens

2. Attribute confusion (Dillon Sheppard):

Attribute	Value	Problem
birth_date	27	Only day number captured
sport	27 february 1979	Full date in wrong field

3. Entity mix-up (Teoctist Arăpașu):

- `birth_place` → `"toader arăpașu"` — actually his **birth name**
- `birth_date` → `"february 7, 1915 – july 30, 2007"` — birth+death merged

Challenges & Limitations

Challenge	Details
WikiBio tokenization	Special tokens (`-lrb-`, `-rrb-`) needed custom cleanup
Multi-sentence coreference	Pronoun → title heuristic misses family member references
Multi-valued attributes	Years, clubs, stats as lists are hard to linearize
No intermediate ground truth	Cannot evaluate the KG directly, only the final table
Evaluation gap	chrF and fuzzy F1 don't capture wrong attribute assignment
Long input degeneration	Finetuned REBEL repeats tokens on long biographies
Entity resolution noise	Hybrid pipeline can produce `"new york new york"` dup objects

Conclusion & Future Work

Achievements:

- Modular **paragraph → knowledge graph → table** system for biographies
- Three extraction approaches: REBEL-only, GLiNER + REBEL, BERT + LUKE + SBERT
- BERT + LUKE: chrF **62.25**, triple F1 **0.671** on WikiBio test set (34,854 records)

TODO: Update with REBEL/hybrid results once evaluation completes

Future Work:

- Better coreference resolution (beyond pronoun → title heuristic)
- Cross-domain generalization beyond biographies
- End-to-end joint training (text → table without separate stages)
- Improved handling of multi-valued attributes (years, stats)

References

- Huguet Cabot & Navigli (2021) — *REBEL: Relation Extraction By End-to-end Language generation*
- Zaratiana et al. (2023) — *GLiNER: Generalist Model for Named Entity Recognition*
- Yamada et al. (2020) — *LUKE: Deep Contextualized Entity Representations*
- Reimers & Gurevych (2019) — *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*
- Lebret et al. (2016) — *Neural Text Generation from Structured Data (WikiBio)*
- Wu et al. (2022) — *Text-to-Table: A New Way of Information Extraction*

Fine-tuned REBEL model: *TOMATOsJr/Rebel_Finetuned_on_Triples* (HuggingFace)